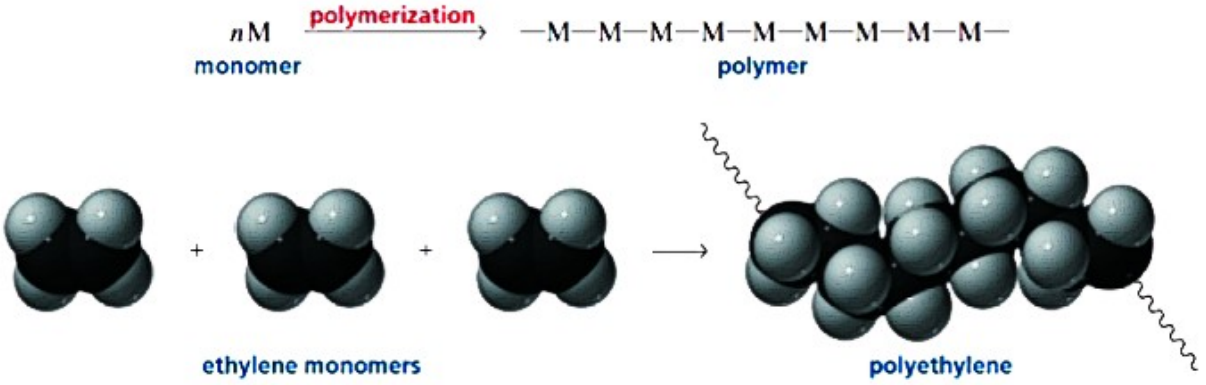
Polymer

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Introduction and Classification

Probably no group of synthetic compounds is of great significance to present-day life than synthetic polymers. Polymers are large molecules that are made by linking together many small molecules .These small molecules are called monomers. The process of linking them together is called polymerization.



In contrast to small organic molecules, which are of interest because of their *chemical* properties, these macromolecules are fascinating because of their *physical* properties, which make them handy in everyday life. Some synthetic polymers look like natural substances, but most are quite distinct from those found in nature. Such diverse items as compact discs, photographic film, food wrap, rugs, artificial joints, bottles, super glue, toys, automobile body parts, condoms , weather stripping, and shoe soles are made of synthetic polymers.

Polymers can be classified into two broad groups: **synthetic polymers** and **biopolymers** (natural polymers). Synthetic polymers are prepared by scientists, whereas biopolymers are synthesized by organisms. RNA and proteins, the molecules that induce biochemical transformations; Deoxyribonucleic acid, the molecule that carries genetic information—the molecule that determines whether a fertilized egg becomes a honeybee or a human; and polysaccharides are examples of biopolymers. The structures and properties of these molecules are presented in other chapters. In this chapter, we will explore synthetic polymers.

Humans first relied on naturally occurring polymers for attire, covering themselves in creature skins and furs. Afterward, they figured out how to spin natural fibers into thread and to weave the thread into cloth. Today, much of our clothing is made of *synthetic polymers* (e.g., nylon, polyester, polyacrylonitrile). Many people prefer clothing made of natural polymers (e.g., cotton, wool, silk), but it has been estimated that if man-made polymers were not accessible, all the arable land in the United States would have to be used for the production of cotton and fleece for apparel.

The first **plastic**—a polymer fit for being molded—was celluloid. It was invented in 1856 by Alexander Parke, it was combination of camphor and nitrocellulose. Celluloid was utilized in the making of piano keys and billiard balls, replacing scarce ivory. The invention of celluloid provided a a relief to numerous elephants, but caused some moments of distress in billiard parlors because nitrocellulose is flammable as well as explosive. Celluloid was utilized for motion picture film until it was replaced by cellulose acetate, a less hazardous polymer.

The first man-made fiber was rayon. In 1865, an epidemic threatened the French silk industry that killed many silkworms, highlighting the requirement for an artificial silk substitute. Louis Chardonnet coincidentally found the beginning material for a synthetic fiber when, while cleaning up some spilled nitrocellulose from a table, he saw long silklike strands adhering to both the table and the cloth. This amazing "Chardonnet silk" was presented at the Paris Exposition in 1891. It was called *rayon since it was sparkling to the point that it seemed to radiate beams of light. The first synthetic rubber was prepared by German chemists in 1917. Their endeavors were in response to a severe shortage of crude materials as a result of barricading during World War I.*

Hermann Staudinger was quick to perceive that the different polymers being synthesized were not disorderly conglomerates of monomers, but were comprised of chains of monomers joined together. Today, the synthesis of polymers has developed from a process carried out with very less chemical understanding to a sophisticated science in which molecules are designed with predetermined descriptions in order to make new materials custom fitted to fit human requirements. Recent illustrations of the many new polymers that are constantly being engineered include Dyneema, the strongest fabric commercially available, and Lycra, a fabric with elastic properties.

Polymer science is a part of the larger discipline of materials science, which includes the formation of new materials to replace ceramics,metals, glass, cardboard, wood, paper; and fabrics. Polymer chemistry has advanced into a multibillion-dollar industry. Currently, there are about 30,000 patented polymers in the United States. More than kilograms of synthetic polymers are created in the United States each year, and we can predict that many more new materials to be developed by scientists in the years to come.

→ The term polymer is used to depict a very large comprised that is made up of many small replicating molecular units. These small repeating molecular units from which the polymer is formed are known as monomers.

→ The chemical reaction that joins the monomers together is called polymerization.

→ Starting from n molecules of a compound S, linking in a linear manner will form polymer $x - S - (S)_{n-2} - S - y$. The nature of linkages at the terminals units i.e. S– x and S – y relies upon the mode of reaction used in making the polymers.

ON THE BASIS OF DIFFERENT UNITS OF MONOMERS, POLYMERS ARE OF TWO TYPES :

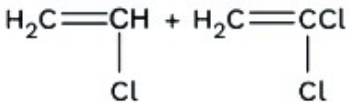
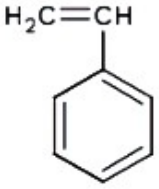
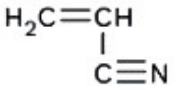
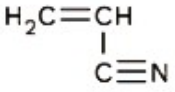
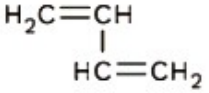
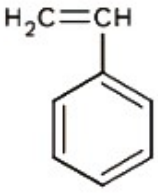
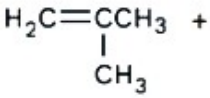
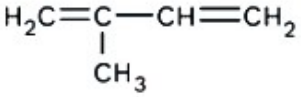
1. **Homopolymer**

Homopolymer is a polymers which consists only one sort of monomer units.
Examples:

HOMOPOLYMER	MONOMER
Starch	Glucose
Cellulose	Glucose
Glycogen	Glucose
Dextrin	Glucose
Inulin	Fructose
Polyethylene	Ethylene
Polyvinyl chloride	Vinyl chloride
Teflon	Tetrafluoro ethylene
Nylon–6	Caprolactam
Polystyrene	Styrene
Orlon (Acrilan)	Acrylonitrile
Plexiglass (Lucite)	Methyl methacrylate
Polyvinyl acetate	Vinyl acetate

2. Copolymer

Copolymer is a polymers, which consists more than one sort of monomeric units.
Examples:

MONOMER	COPOLYMER NAME	USES
 vinyl chloride + vinylidene chloride	Saran	film for wrapping food
 styrene +  acrylonitrile	SAN	Dishwasher-safe objects, vacuum cleaner parts
 acrylonitrile +  1,3-butadiene +  styrene	ABS	bumpers crash helmets, telephones, luggage
 isobutylene +  isoprene	butyl rubber	inner tubes, balls, inflatable sporting goods

Types of Copolymers

There are four types of copolymers on the basis of distribution of repeating unit.

(i) In an alternating copolymer, the two monomeric units distributed in alternating sequence.

(ii) Block copolymer is a copolymer which consists blocks of each kinds of monomeric units.

(iii) Random copolymer is a copolymer in which monomeric units are distributed randomly.

(iv) A copolymer in which branches of one monomeric unit is grafted onto a backbone of another monomeric unit is called graft polymer.

AN ALTERNATING COPOLYMER	ABABABABABABABABABABA
A BLOCK COPOLYMER	AAAAABBBBBAAAAABBBBBAAA
A RANDOM COPOLYMER	AABABABBABAABBABABBAAAB
A GRAFT COPOLYMER	AAAAAAAAAAAAAAAAAAAAAAA
	B B B
	B B B
	B B B
	B B B
	B B B
	B B B

Categorization of Polymers

Ways in which Polymers might be grouped.

(i) By Source

(1) Natural Polymers

Natural polymers are those polymers which are derived from living organisms like plants and animals .

Examples :

Natural Polymer	Monomeric units
1. Silk	Monosaccharide
2. cis polyisoprene (Natural Rubber)	Isoprene (2-Methyl-1,3-butadiene)
3. trans polyisoprene (Gutta percha)	Isoprene

Biopolymers are naturally occurring polymers such as polysaccharides, RNA and DNA which participate in metabolic processes.

2. Semisynthetic Polymers

Semisynthetic polymers are those which are synthesized using naturally occurring polymers. Cellulose is the mostly used raw material to synthesize semisynthetic polymer such as cellulose nitrate, Rayon etc.

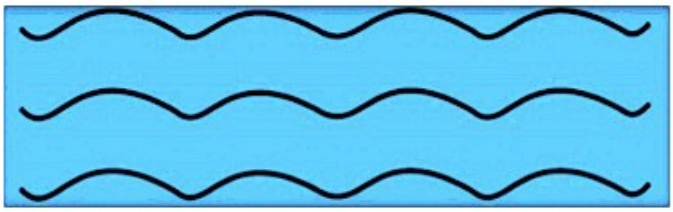
3. Synthetic Polymers

Man-made polymers are those polymers which are synthesized by scientists. Examples: polyethylene, nylon-66, nylon-6, synthetic rubbers etc.

(ii) By Structure

1. Linear Polymers

Linear polymers are polymers consisting long and straight chains. They have well organised structure as these chains are arranged over one another in a straight line.



Due to their tightly packed structure ,they possess high densities and high melting point . They also possess high tensile strength. Conversion of linear polymers into fibres is possible.

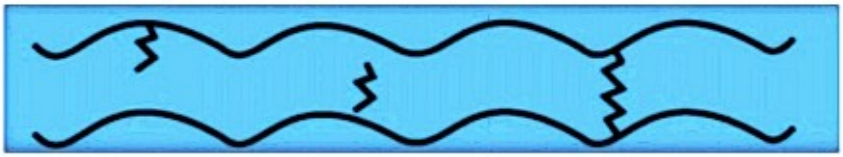
Note :

(i) All fibers are linear polymers. silk, cellulose, nylon, terylene etc. are linear polymers.

(ii) Linear polymers may be addition as well as condensation polymers. Examples are polypeptide, cellulose, nylon, terylene, nucleic acid etc.

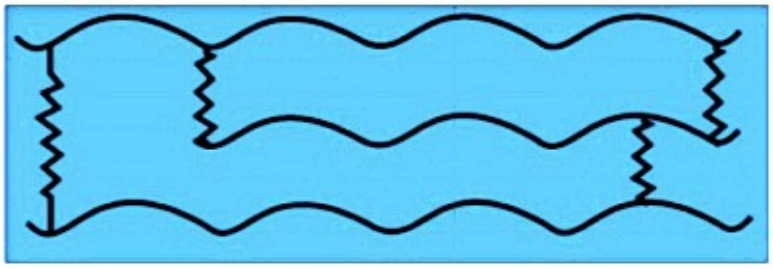
2. Branched Chain Polymers

Branched chain polymers are polymers having branched structure. These may be prepared due to addition as well as condensation polymerization. These polymers have low densities, low melting points, and tensile strength due to their loosely packed structure. Examples are a glycogen, mylopectin, low density polyethylene and all vulcanized rubbers.



3. Cross-linked or Three-Dimensional Network Polymers

Cross-linked polymers are polymer having 3D network structure.This three-dimensional network structure is formed by linking the initially formed linear polymeric chains together. They are always prepared by condensation polymerization. They are hard, rigid and brittle in nature. Example: Resins.



By Intermolecular Forces (Secondary Forces)W

Polymeric chains are held together by intermolecular forces such as

(i) Van der waals forces
(ii) Hydrogen bonds and
(iii) Dipole - dipole interactions.

These secondary forces influence mechanical properties of polymers such as elasticity, toughness, tensile strength etc.. Their Magnitude relies on molecular size and the no. of functional groups along the polymeric chains. As the length of the polymeric chain increases, magnitude of Intermolecular forces also increases. Based on magnitude of intermolecular forces, polymers are grouped into four categories.

1. Elastomers

Polymers having low young's modulus and weak secondary forces are elastomers. They have the ability to regain their original shape after being stretched. This amorphous polymer is randomly aligned. It must have some cross-connects to prevent sliding of chains over one another. In elastomers, weak van der waal's forces are present between polymeric chains.

If stretched or extended, the random chains stretch out, but they are unable to keep them in that arrangement and position due to extremely weak intermolecular forces present between polymeric chains. As soon as the stretching force is removed, they return back to their original arrangement. They possess the ability to extend out over ten times their original length.

Example : vulcanized rubber.

Note : Butadiene and its derivatives undergo addition polymerization to form elastomers (addition polymers).

2. Fibres

They are linear polymers . In fibres ,intermolecular forces present between individual chains of a polymer are dipole-dipole attraction and / or hydrogen bonds. They have well organised structure. They possess high tensile strength, low solubility and high melting points due to strong intermolecular forces and highly ordered geometry. They also have least elasticity . They are of crystalline nature. Examples are wool, silk, terylene etc.

Note :

(i) Condensation polymers prepared from monomeric units having two functional groups are fibres in character.

(ii) Addition polymers of alkene derivatives having strong –I group are fibres in character.

3. Thermoplastic Polymers

Thermoplastics are polymers consisting both ordered ,crystalline regions and amorphous ,non-crystalline areas. There are no cross-connects between the polymeric chains and intermolecular forces are present in between elastomers and fibres. They are rigid at room temperature. Thermoplastics become soft and viscous on heating as the individual chains can slide over one another and hard on cooling. Thermoplastics can be heated and cooled as many times as required as there will be no change in its mechanical properties and chemical composition. Thermoplastics can be pliable into combs, toys, switch plates, etc.

Examples : polystyrene, polyvinyl chloride, teflon etc.

Note :

Ethylene and ethylene derivatives undergo addition polymerization to form thermoplastic polymers.

4. Thermosetting Polymers

If polymer chains are cross-linked, rigid materials can be obtained. Thermosetting plastic is rigid non recyclable plastic used in computer, TV, Car dashboard etc. Once they are hard remelting is not possible. Bonds are covalent not van der waals. Examples are : phenol formaldehyde resin, urea-formaldehyde resin, melamine - formaldehyde resin.

Difference Between Thermoplastic and Thermosetting Polymers

S.NO.	THERMOPLASTIC POLYMERS	THERMOSETTING POLYMERS
1.	Soften and melt on heating and become rigid on cooling i.e. process is reversible	Become rigid on heating and process is irreversible.
2.	Can be pliable and remoulded and reshaped.	They can be pliable once and cannot be remoulded or reshaped.
3.	They are addition polymers	They are condensation polymers.
4.	Structure is generally linear	Structure is cross - linked.

Q1 **Define thermoplastic and thermosetting polymers. Give one example of each.**

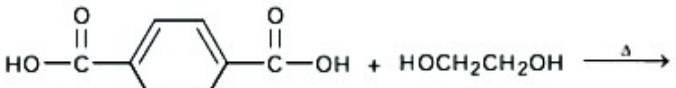
A1 A thermoplastic polymer is a linear or slightly branched polymer that becomes pliable above a particular temperature and reverts to a solid state upon cooling. e.g., Polyethylene, PVC.
Thermosetting polymers are permanently setting polymers on heating in a mould. They get hardened and set and cannot be settled again.
example: Bakelite, Melamine formaldehyde.

Classification Based Upon Synthesis

1. Step-growth polymers

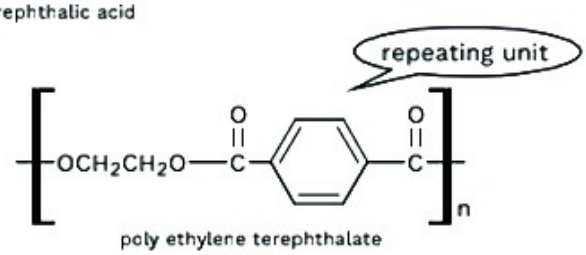
- Step-growth polymers, also called condensation polymers.
- Prepared by condensation reactions i.e., combining molecules having two reactive functional groups and eliminating a small molecule, like water or an alcohol.
- condensation polymerization allows any two reactive molecules to join.
- Monomer units having three functional groups always provide crosslinked polymer.
- Step growth polymer doesn't keep all atoms of monomer units in the polymer.
  Examples : Dacron Urea-formaldehyde resin, phenol-formaldehyde resin.

For Example :



Terephthalic acid

1,2-ethanediol

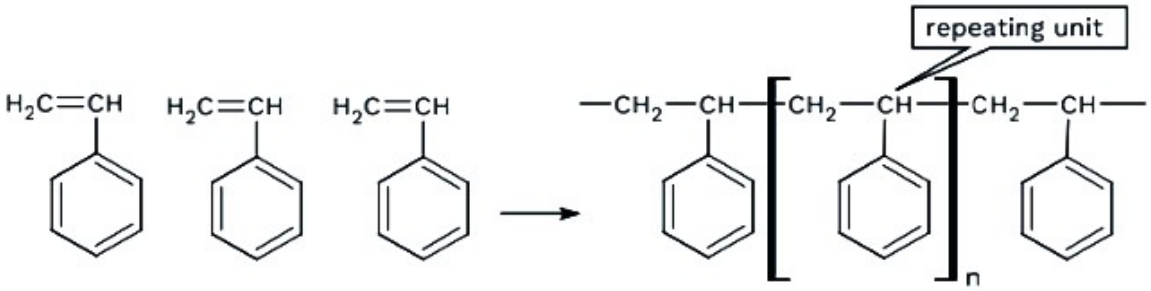


poly ethylene terephthalate
Dacron
a step-growth polymer

2. Addition polymerisation

- Chain-growth polymers, also termed as addition polymers,
- Prepared by addition reactions i.e., individual molecules are added to the end of developing chain.
- The steps involve in chain growth polymerization are initiation, chain propagation and chain termination.
- Terminal of the chain i.e. ,cation, an anion ,or a radical is reactive.
- Addition polymers constitute all the atoms initially present in the monomer units.

Examples: Polystyrene (utilized for disposable food containers, toothbrush handles, insulation), polytetrafluoroethylene, polyacrylonitrile etc.

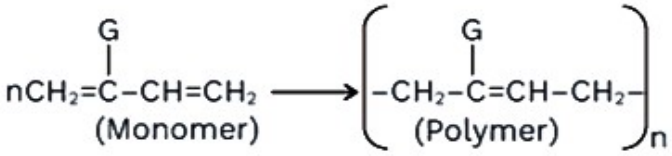


styrene

polystyrene
a chain-growth polymer

- Polymer is always branched chain for monomeric unit 1, 3-butadiene or

2-substituted-1, 3-butadiene $\left(CH_2 = \underset{\underset{G}{|}}{C} - CH = CH_2\right)$,



(Repeating unit)	(Polymer)
(i) G = Cl ; 2 -Chloro-1,3-butadiene or chloroprene	(i) Polychloroprene (Neoprene)
(ii) G = H : 1,3 - Butadiene	(ii) Polybutadiene
(iii) G = CH_3 ; 2 - Methyl-1,3-butadiene or isoprene	(iii) Polyisoprene

Q2 **Differentiate between condensation and addition polymers based on mode of polymerization. Give one example of each type.**

A2 **Addition polymers :** The polymers formed by the direct addition of a large number of monomers units are called addition polymers. For example polystyrene, polythene.

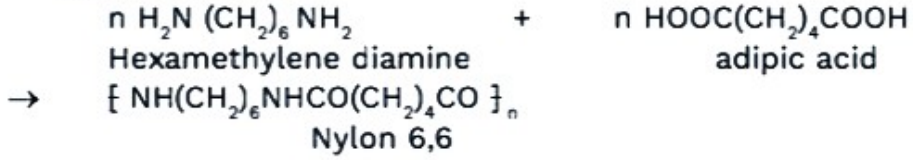
Condensation polymers : The polymers prepared by condensation reaction between two or more than two monomers by the loss of small molecules such as H2O, alcohol etc. are called condensation polymers. For example, nylon-66, bakelite.

Q3 **What is step growth polymerisation?**

A3 **Step growth polymerisation :** The polymerisation which progresses through step by step is called step growth polymerisation. It is also known as condensation polymerisation. Nylon and bakelite are examples of condensation polymers.

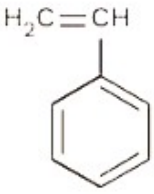
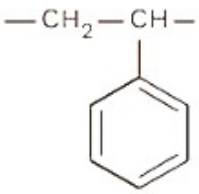
These condensation reactions may results in loss of simple molecules such as water, alcohol etc.

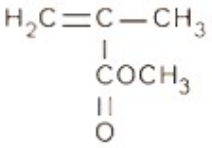
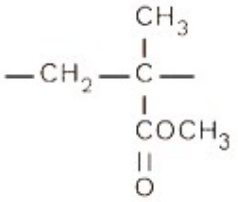
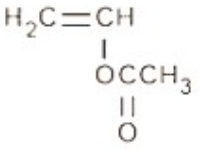
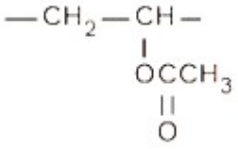
Example : Nylon 6, 6 is prepared by the condensation of hexamethylene diamine with adipic acid.



Mechanism Based Polymers

Some Important Chain-Growth Polymers and their uses

Monomer	Repeating unit	Polymer name	Uses
$CH_2=CH_2$	$-CH_2-CH_2-$	polyethylene	film, toys, bottles, plastic bags
$H_2C=CH$ (Cl)	$-CH_2-CH-$ (Cl)	polyvinyl chloride	"squeeze" bottles, pipe, siding, flooring
$CH_2= CH - CH_3$	$-CH_2-CH-$ (CH_3)	Polypropylene	molded caps, margarine tubs, indoor/outdoor carpeting, upholstery
		polystyrene	toys, packaging, clear cups, egg cartons, hot drink cups
$CF_2 = CF_2$	$- CF_2 - CF_2 -$	poly tetrafluoroethylene (Teflon)	nonsticking surfaces, liners, cable insulation
$H_2C=CH$ ($C\equiv N$)	$-CH_2-CH-$ ($C\equiv N$)	polyacrylonitrile (Orlon; Acrilan)	rugs, blankets, yarn, apparel, simulated fur

		polymethyl-methacrylate (plexiglass, lucite)	lighting fixtures, signs, solar panels, skylights
		polyvinyl acetate	latex paints, adhesives

Addition polymers keep all the atoms of the monomer units in the polymer. Addition polymerization takes place in three steps: Initiation, chain propagation and chain termination.

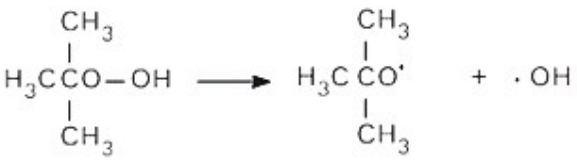
Note :
Ethylene and substituted ethylenes are the most common monomeric units used in addition polymerization to form vinyl polymers. Alpha olefins are monosubstituted ethylenes.

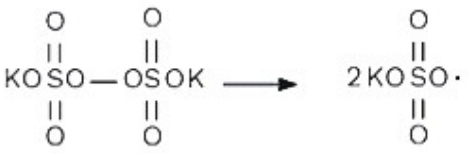
Types of Addition Polymerization

(A) Radical Polymerization :

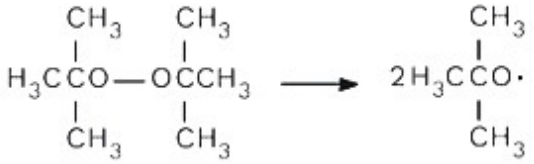
It occur in the presence of free radical generating initiators such as benzoyl peroxide, tert-butyl peroxide etc.,

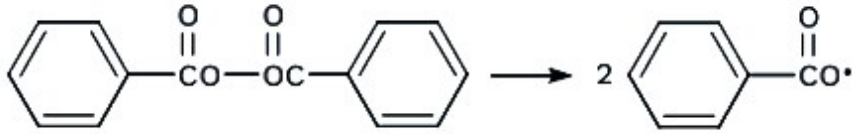
Some initiators

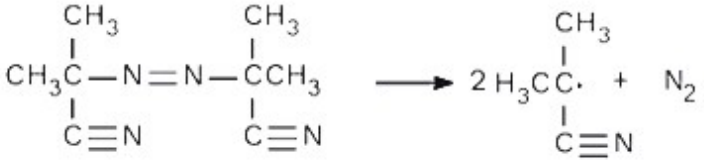
1. 

2. 

Reaction intermediate of radical polymerization is a free radical. Radical polymerization has more possibility for those monomers whose free radicals are more stable.

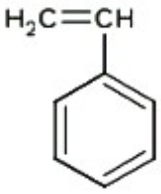
3. 

4. 

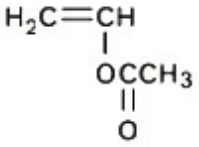
5. 

Radical polymerization involves the formation of free radical intermediate .Monomer units whose intermediates are more stable has more chances to undergo radical polymerization.

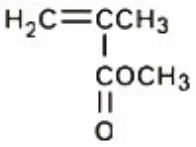
Examples of Alkenes that Undergo Radical Polymerization



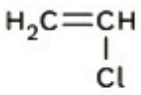
styrene



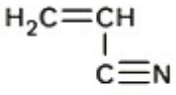
vinyl acetate



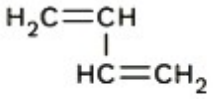
methyl methacrylate



vinyl chloride

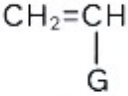


acrylonitrile

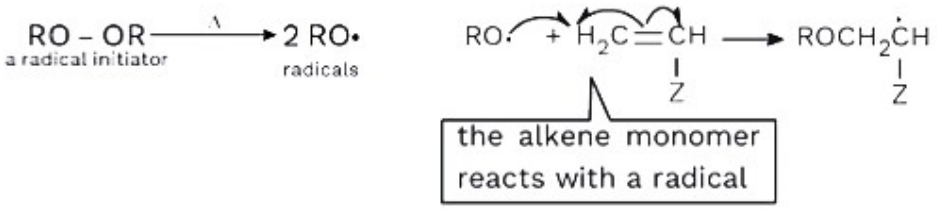


1,3-butadiene

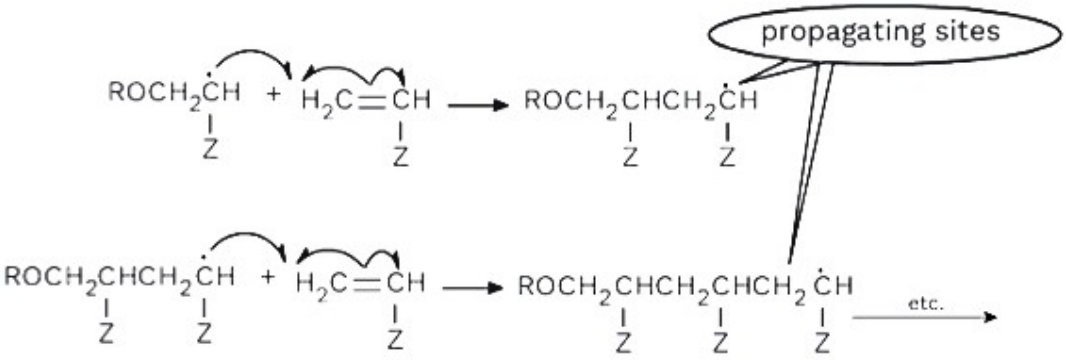
Radical polymer has straight as well as branched chain structure.

Most of the commercial chain growth polymers are vinyl polymers obtained from alkenes and their derivatives  [G is H, C_6H_5,R etc.]. This sort of polymerization is performed by heating the monomer unit only a very less amount of the initiator or by exposing the monomer unit to light. The general mode of radical polymerization of vinyl monomers is depicted below :

Chain-initiating steps



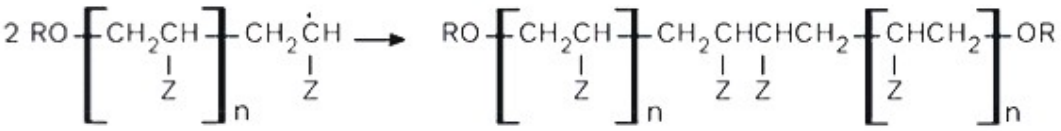
Chain-propagating steps



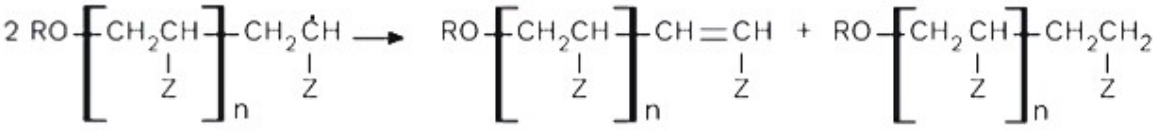
This process is repeated again and again.

Three ways to terminate the chain

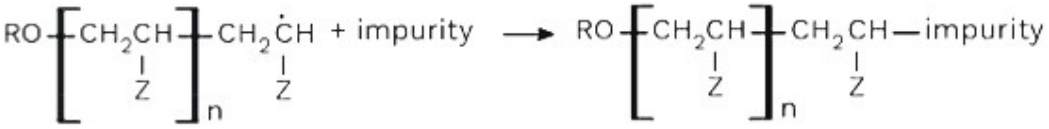
Chain Combination



Disproportionation

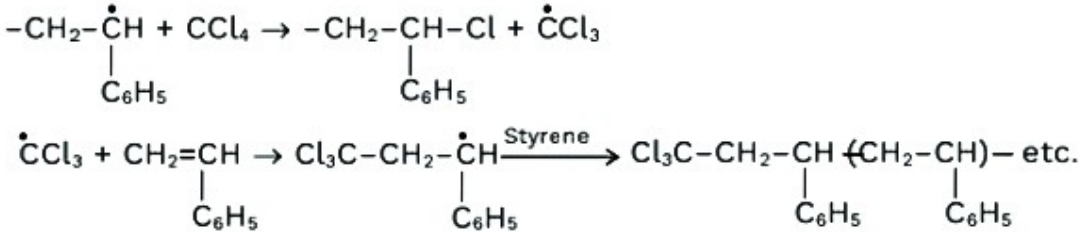


Reaction with an impurity



For example :

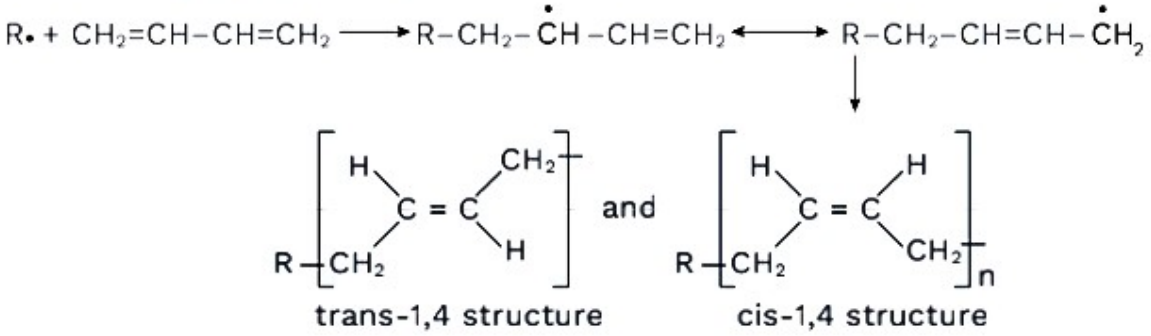
In the presence of CCl_4, styrene polymerizes to form polystyrene of a lower average molecular mass which also have some chlorine, what happens here is that growing polystyrene radical which normally would add on a monomer reacts with the chain transfer agent to end the original chain and give rise to a new radical. The latter starts a new polymerization chain and thereby forms a new polymer as depicted below.



Types of polymerization possible for diene ::

1. 1,4-polymerization.

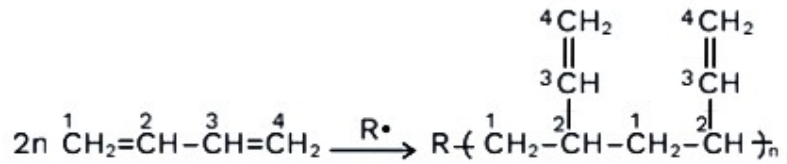
1,3-butadiene undergo polymerization at C_1 and C_4 to form unbranched polymer which is different from that formed from an alkene which at each of its carbons is substituted by unlike groups and even exist either as trans-polybutadiene or cis polybutadiene or a mixture



trans-1,4 structure cis-1,4 structure

2. 1,2- polymerization

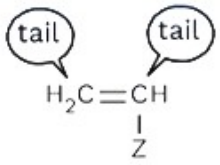
Polymeric product , polyvinyl polythene is formed when 1,3 -butadiene undergo polymerization at C_1 and C_2.

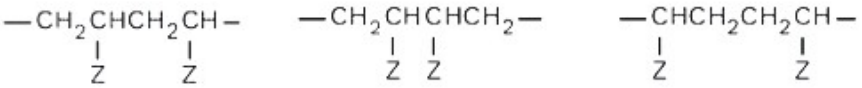


In order to modify the properties of polymers, Initial polymers are treated with chemicals

to connect their double bonds. Formation of rubber is based on these reactions.

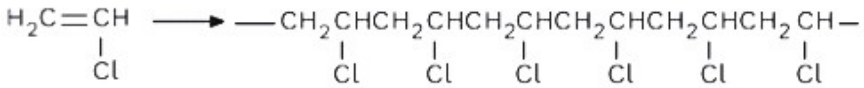
Note :
Monosubstituted ethylenes undergo head-to-tail addition polymerization, where the head of one monomer unit is attached to the tail of another monomer.





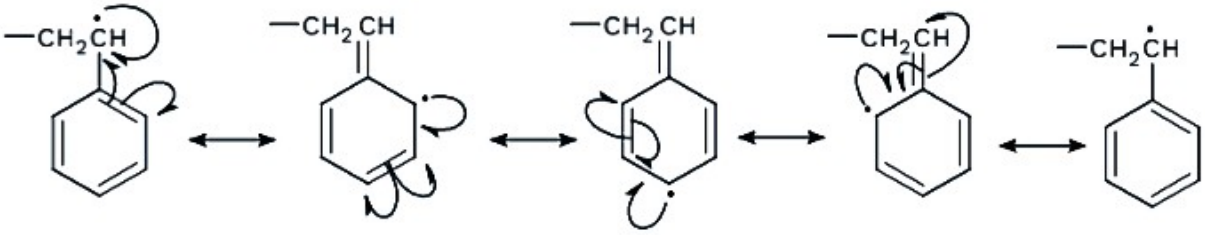
head-to-tail head-to-head tail-to-tail

Head-to-tail addition of substituted ethylene results in a polymer in which every other carbon bears a substituent.



vinyl chloride poly(vinyl chloride)

In Head-to-tail addition ,less sterically hindered unsubstituted carbon of alkene is attacked by the propagating site. Head-to-tail addition is also preferred by radical stablising groups. For instance, if Z is a phenyl substituent, then the benzene ring stabilizes the radical by electron delocalization. Therefore the carbon that bears the phenyl substituent is the propagating site.



Q1 **The monomer that undergo radical polymerisation most easily is**

(A) $CH_2=CH_2$ **(B)** $C_6H_5CH=CH_2$

(C) $H_2C=C(Me)(Me)$ **(D)** $CH_3-CH=CH_2$

A1 (B)

B. Cationic Polymerization :

Some important points related to cationic polymerization :

1. Monomers that are best able to go through polymerization by a cationic mechanism are those with electron - donating substituents that can stabilise the carbocation.
2. initiated by an electrophile
3. Reaction intermediate is carbocation.
4. Rearrangement in case of more stable carbocation formation.
5. Most common electrophile is $BF_3.OEt_2$.
6. Chain termination step completed with the help of base(electron donating group.

$CH_2=CH-CH_3$ Propene

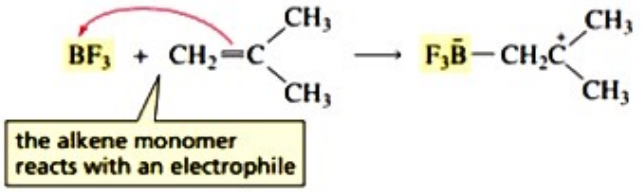
$CH_2=C(CH_3)-CH_3$ Isobutene

$CH_2=CH-O-C(=O)-CH_3$ Vinyl acetate

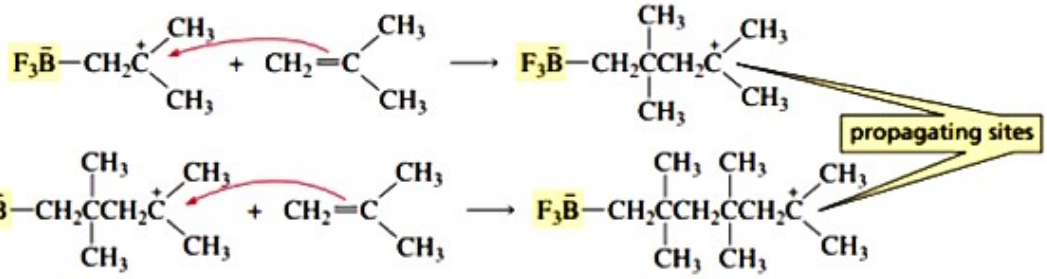
$C_6H_5-CH=CH_2$ Styrene

cationic polymerisation steps below.

chain-initiating step



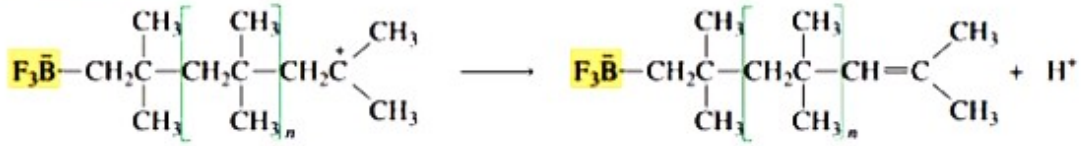
chain-propagating steps



Cationic polymerization is terminated by removing a proton or by adding a by reacting with solvent or nucleophile that reacts with the propagating site by reacting with solvent..

three ways to terminate the chain

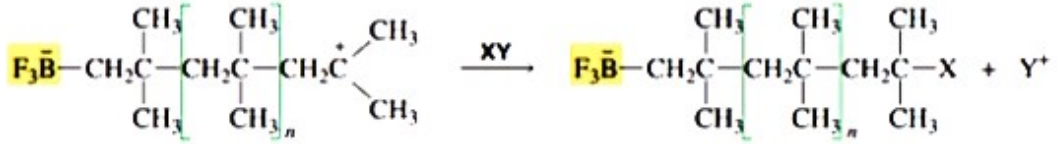
loss of a proton



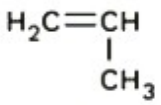
reaction with a nucleophile



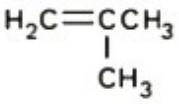
chain-transfer reaction with the solvent



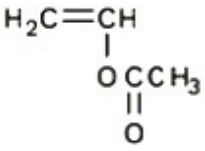
Examples of Alkenes that Undergo Cationic Polymerization:



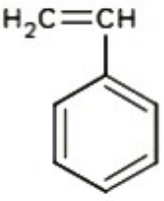
propylene



isobutylene

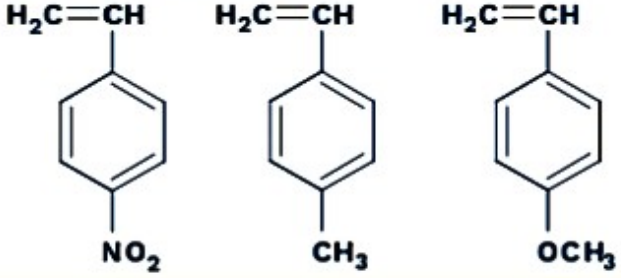


vinyl acetate

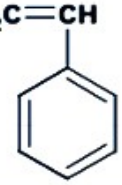
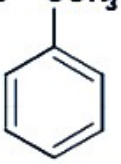


styrene

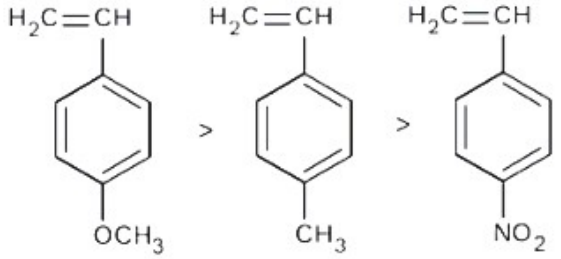
Q2 **List the following groups of monomers in order of decreasing ability to undergo cationic polymerization:**

(A) 

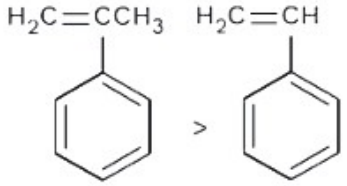
(B) $H_2C{=}CHCH_3$ $H_2C{=}CHOCCH_3$ (O) $H_2C{=}CHCOCH_3$ (O)

(C) $H_2C{=}CH$  $H_2C{=}CCH_3$ 

A2

(A) 

(B) $H_2C{=}CHOCCH_3$ (O) > $H_2C{=}CHCOCH_3$ (O) > $H_2C{=}CHCH_3$

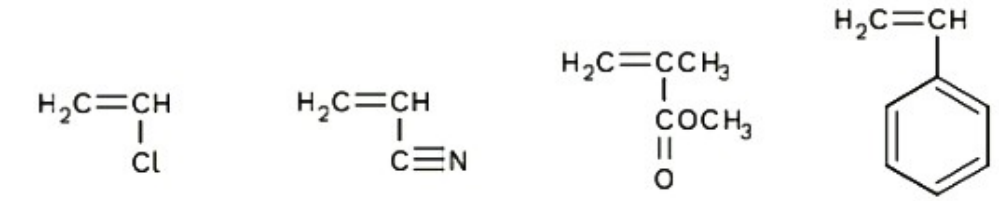
(C) 

Stable carbocation is the basis of cation polymerization.

C. Anionic Polymerization

- This polymerization involves the initiator attack on an alkene to form an anion i.e, propagating site but this attack does not occur quickly because of electron rich alkenes .
- The alkene must contain an electron-withdrawing substituent to decrease its electron density
- Initiator must be a very good nucleophile or base like butyllithium or sodium amide. The suitable initiator are NaNH2 or RLi.

Examples of Alkenes that Undergo Anionic Polymerization:

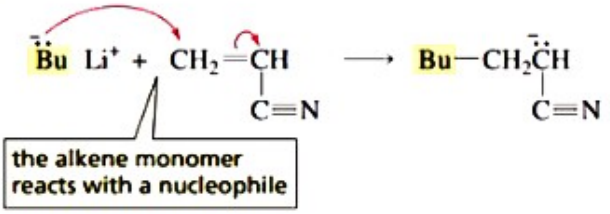


vinyl chloride | acrylonitrile | methyl methacrylate | styrene

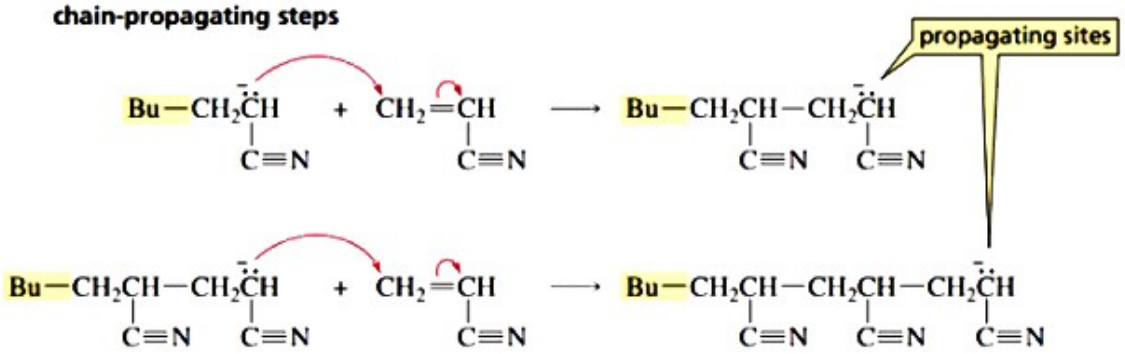
- Linear polymer is formed as product in anionic polymerization.
- In anionic polymerization, an acid is used to terminate the chain.

Example : Styrene undergo anionic polymerization in presence of potassium amide to form polystyrene. Steps involved in anionic polymerization are as follows:

chain-initiating step

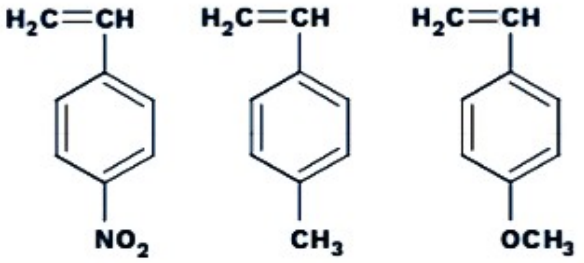


chain-propagating steps

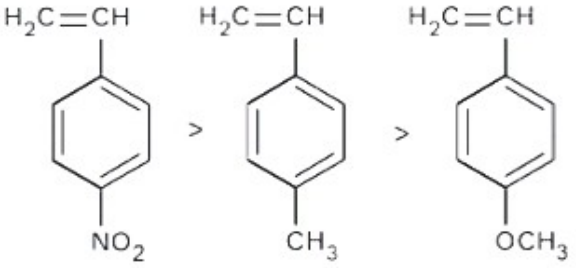


The chain can be terminated by a chain transfer reaction with the solvent or by reaction with an impurity in the reaction mixture. If the chain is not terminated, then chain propagation will not stop until all the monomer has been devoured. Further addition of monomer leads to continuation of polymerization reaction Because of active propagating site. Polymers having such non terminated chains are called Living polymers as these chains remain active while their degree of termination is still negligible. These polymers are formed from anionic polymerization because the chain termination cannot be done by proton loss from the polymer.

Q3 **List the following groups of monomers in order of decreasing ability to undergo anionic polymerization:**



A3



Stable carbanion is the basis of anion polymerization.

Q4 **What kind of polymerisation, cationic or anionic or vinyl polymerisation, would be suited for the preparation of**

(A) Polypropylene
(B) Poly acrylonitrile
(C) Polystyrene

A4 (A) Polypropylene → Cationic Polymerisation
(B) Poly acrylonitrile → Anionic Polymerisation or Radical Polymerisation
(C) Polystyrene → Cationic Polymerisation

Q5 **Choose addition polymer among the following**

(A) nylon 6,6 **(B) nylon 6**
(C) Buna–N **(D) terylene**

A5 C

D. Ziegler- Natta polymerization :

Addition polymerization which takes place in the presence of Ziegler- Natta catalyst [$(C_2H_5)_3$ Al and $TiCl_4$] is known as Ziegler- Natta polymerization or coordination polymersation. Ziegler- Natta polymerization always gives linear, stereo-regular polymers. Ziegler- Natta catalyst revolutionized the field of polymer chemistry because they allow the synthesis of stronger and stiffer polymers (due to linear geometry) that have greater resistance to cracking and heat. High density polyethylene is prepared using a Ziegler- Natta catalyst.

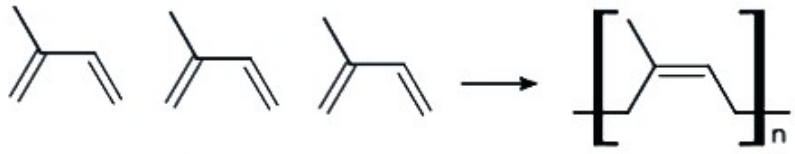
Rubber and Other Polymers

Natural Rubber

Natural rubber is mainly obtained from latex of braziliensis tree. But, can also be extracted from around five hundred different plants. This latex is coagulated with formic acid and acetic acid and the coagulated mass formed is then squeezed.

The raw natural rubber

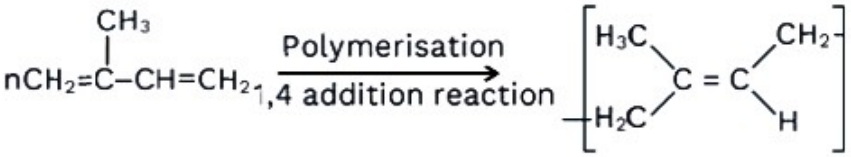
- is a mushy gummy, and sticky mass which dissolves non-polar solvents.
- have low tensile strength and low elasticity.
- is a polymer of isoprene (2-methyl-1,3-butadiene).Nearly 5000 isoprene molecules combine by head to tail to form rubber.
- Consists only cis double bonds



isoprene units

cis-poly(2-methyl-1,3-butadiene)
natural rubber

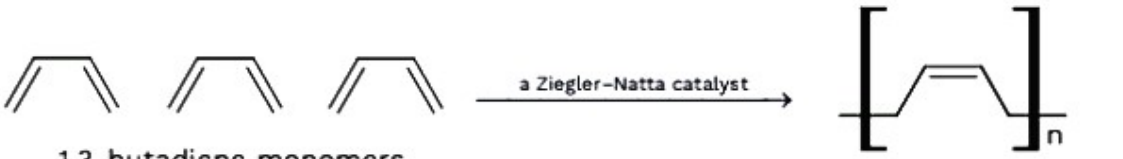
Gutta-percha is a naturally occuring polymer. The term "gutta percha" is a Malaysian words getah, means "gum," and percha, means "tree" . It is rigid, brittle latex that isderived from certain trees such as Palaquim gutta. It is also a polymer of isoprene but double present in it are trans.



Synthetic Rubbers

Some points related to Synthetic rubber.

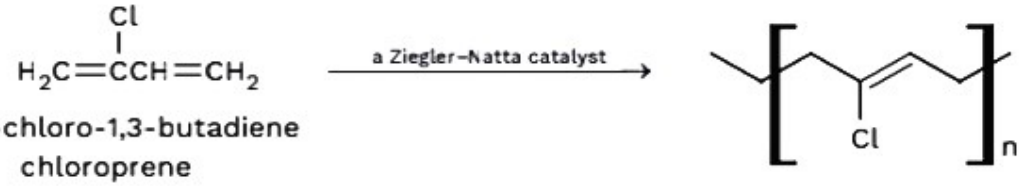
- made by polymerizing dienes except isoprene in the presence of a Ziegler–Natta catalyst to control the configuration of double bonds in polymer.
- One such synthetic rubber is cis-poly(1,3-butadiene) which consists only cis double bonds.



1,3-butadiene monomers

cis-poly(1,3-butadiene)
a synthetic rubber

- A polymer of 2-chloro-1,3-butadiene i.e Neoprene is also a synthetic rubber but consisting only trans double bonds. Shoe soles, tires, hoses etc are made by using this polymer.

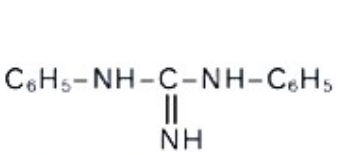


2-chloro-1,3-butadiene
chloroprene

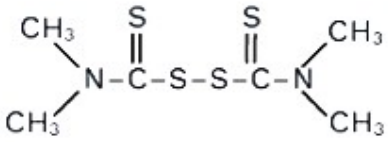
neoprene

Alssue related to rubbers i.e., natural and synthetic rubbers is their polymers soft and sticky nature. Cab This issue can be solved by Vulcanization ,which was discovered by Charles Goodyear .He accidently dropped the mixture of rubber and sulphur on a hot stove and found that the mixture turned hard but still flexible. Vulcanisation :

- a process in which rubber is heated with sulphur or sulphur containing compound at 150° C for few hours.
- involves the formation of cross-linking between the polymeric chains through disulphide bonds as this increases mechanical strength of rubber.
- upgraded by using certain organic compounds known as accelerators such as :

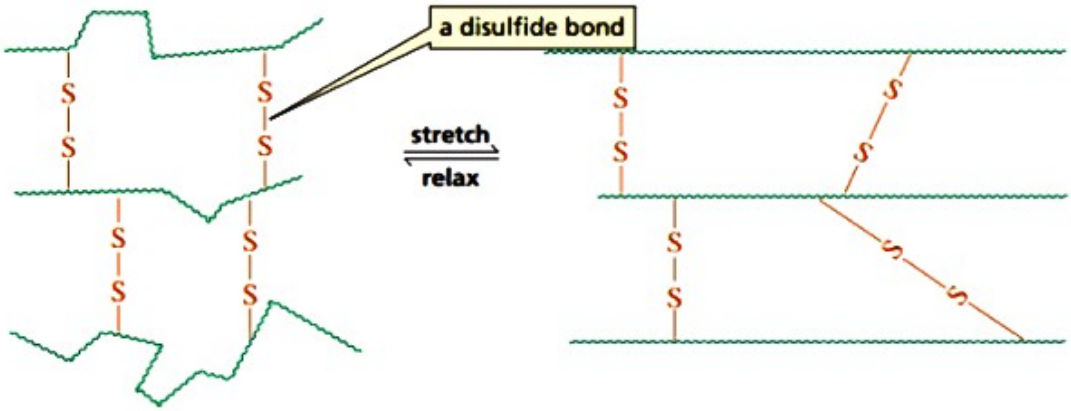


Diphenyl guanidine



Tetramethylthiuramdisulphide

Instead of the individual chains just being entangled together, the vulcanized chains are covalently bonded together in one giant molecule.



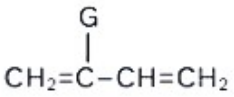
- Fillers like zinc oxide ,carbon black etc., are added to the crude rubber before its vulcanising it to enhance its wearing characteristics. Water-proof coats, shoes etc., are made by using natural rubber whereas gloves tubing, car tyres etc., are made by using vulcanised rubber.

Synthetic Rubber or Polymerisation of Dienes

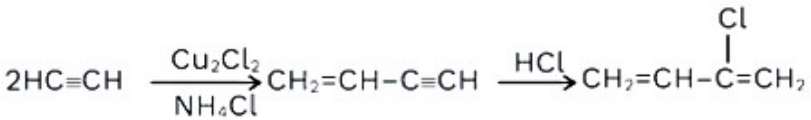
Polymers of 1,3 butadienes are known as synthetic or man- made rubber .Man made rubbers possess enhanced properties such as flexibility, toughness and durability as well as some properties of natural rubber like elasticity, water proof etc.,

1. Homopolymers

2-substituted - 1,3- butadienes is the monomeric unit of this category..

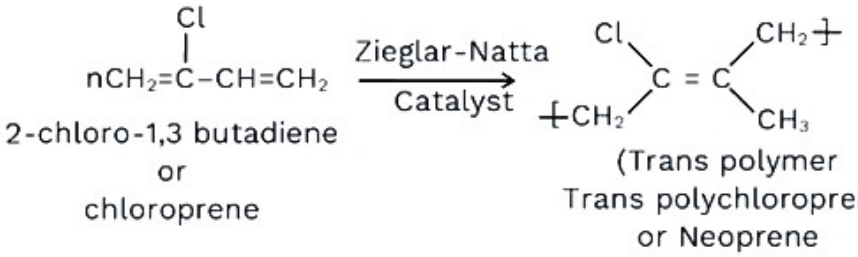


where G=H, CH_3 or Cl.



Acetylene Vinyl acetylene Chloroprene

Free radical polymerisation of chloroprene results in the formation of polychloroprene (neoprene).



2-chloro-1,3 butadiene
or
chloroprene

(Trans polymer
Trans polychloroprene
or Neoprene

Most of the properties of neoprene are similar to that of natural rubber, but it is more impervious to activity of gasoline, oils and other hydrocarbons , sunlight, oxygen, ozone and heat. It is non - inflammable. Automobile parts, hoses for oil containers, insulation of electric wires etc., are made up of neoprene.

Q1 **Based on molecular forces what type of polymer is neoprene?**

A1 Elastomer (weakest intermolecular forces)

2. Copolymers

Examples:

Synthetic rubber	**Monomers**
1. Buna-S, SBR (Styrene-butadiene rubber)	$C_6H_5–CH=CH_2$ (25%) + $CH_2=CH–CH=CH_2$ (75%)
2. Buna-N, NBR (Nitrile-butadiene rubber)	$CH_2=CH–CN$ (25%) + $CH_2=CH–CH=CH_2$ (75%)
3. Butyl rubber	$CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2$ 98% + Butadiene 2%
4. ABS; Acrylonitrile, Butadiene, Styrene	$CH_2=CH–CN$ + $CH_2=CH–CH=CH_2$ + $C_6H_5CH=CH_2$

(a) Thiokol :

Ethylene chloride and sodium polysulphide are polymerised to form Thiokol.

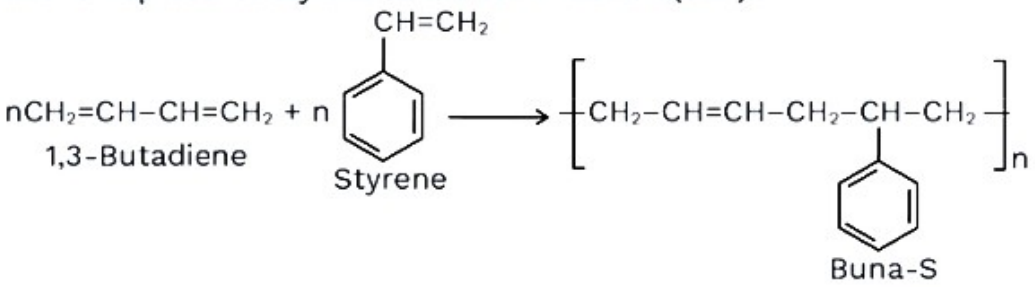
$$2Cl–CH_2–CH_2–Cl + Na–S–S–Na \xrightarrow{\text{Polymerisation}} –CH_2–CH_2–S—S–CH_2–CH_2–S—S–CH_2–CH_2–$$

Thiokol rubber

The unit that repeats in Thiokol rubber is $–CH_2–S–S–CH_2–$.Chemically resistant polymer, Thiokol is utilized in manufacturing engine gaskets, rocket fuel etc., on mixing with oxidizing agents.

(b) Buna -S (SBR : Styrene-butadiene rubber) :

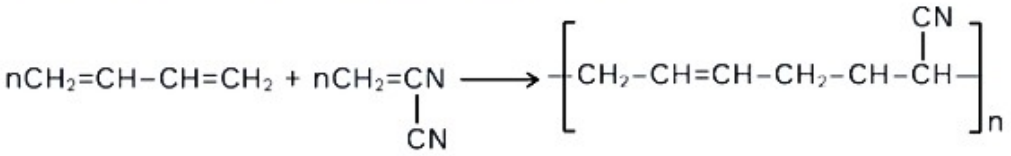
It is also known as GRS or General purpose styrene Rubber .It is an elastomer formed by free radical copolymerisation of styrene and butadiene in presence of sodium metal. 'Bu' represents butadiene, 'na' represents symbol of sodium and 'S' represents styrene in the term "Buna-S(SBR)".



In Buna-S ,carbon black is used as filler before vulcanising it with sulphur. Tyres and other mechanical rubber products are manufactured using Buna-S due to wear and tear resistant.

(c) Buna-N :

Copolymerising acrylonitrile (General Purpose Rubber acrylonitrile or GRA) and butadiene results in formation of Buna -N.



Rigid Fuel tanks are made using Buna -N as it is resistant to the action of lubricating oil, petrol etc.,

(d) Cold Rubber :

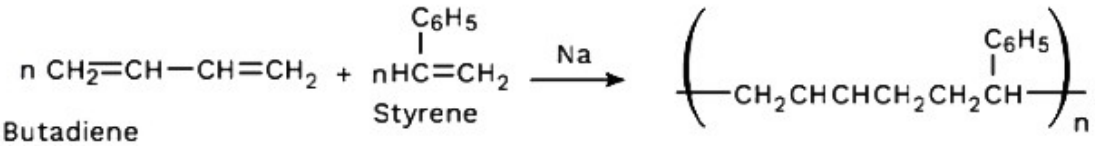
At –18° to 5°C temperature , styrene and butadiene are polymerised in the presence of a redox system to form cold rubber. Unlike SBR, Cold rubber shows more resistance to abrasion .It possess more tensile strength.

Q2 **What are elastomers? Give the chemical equation for preparation of Buna-S.**

A2 **Elastomers :** Those polymers which can be stretched by applying an external force but on removal of the force such polymers regain their position are called elastomers. These polymers are made up of long carbon chains, cross linked with weak forces. For example rubber.

Preparation of Buna-S : Buna-S is a synthetic rubber, prepared by polymerising

butadiene and *styrene* together.



Butadiene

Styrene

Styrene-butadiene rubber or Butadiene-styrene copolymer Buna-S)

Q3 **Write the structures of monomers utilized and one use of each of the following polymers.**
(a) Teflon **(b) Buna-N**

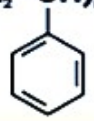
A3

Polymer	Monomer	Use
(a) Teflon	$CF_2 = CF_2$ Tetrafluroethylene	In coating utensils to make them non-sticking.
(b) Buna-N	$CH_2 = CH-CH = CH_2$ and $HC(CN)=CH_2$	In making of seals and tank linings

Q4 **Write the name and structure of the monomer of Buna-S:**

A4 **Buna-S :** Monomer are 1, 3–butadiene and styrene
$CH_2 = CH - CH = CH_2$ and $C_6H_5 - CH = CH_2$

Q5 **If $-(CH_2-CH)_n-$ a homopolymer or a copolymer?**



A5 Homo polymer

Q6 **Distinguish between homopolymers and copolymers and give an example of each class.**

A6 **Homopolymers :** Polymers which consists units only one sort of monomer unit are called homopolymers.
Examples : PVC, nylon 6 etc.
Copolymers : Polymers which consists two or more sort of monomer units are called copolymers.
Examples : Bakelite, Buna-S, and nylon 66 etc.

Nylon

All man -made fibres composed of polyamides are termed as Nylon. The number of carbon atoms present in the monomeric units are represented by putting a suffix with the term Nylon.

1. Nylon – 6, 6 (Nylon six, six)

Adipic acid (a dibasic acid having 6 carbon atoms) and hexamethylenediamine (a diamine with six carbon atoms) undergo condensation polymerization to form Nylon-6,6 as product..

$$n\underset{\text{Adipic acid}}{HOOC(CH_2)_4COOH} + n\underset{\text{Hexamethylene diamine}}{H_2N(CH_2)_6NH_2} \xrightarrow[\text{High pressure}]{280^{\circ}C} \underset{\text{Nylon – 6, 6}}{\left[OC(CH_2)_4CONH(CH_2)_6NH\right]_n} + (n-1)\,H_2O$$

2. Nyon-6, 10 (Nylon six, ten)

ISebacic acid (a dibasic acid with 10 carbon atoms) and hexamethylenediamine (six carbon atoms) undergo condensation polymerisation to give Nylon-6,10 as product.

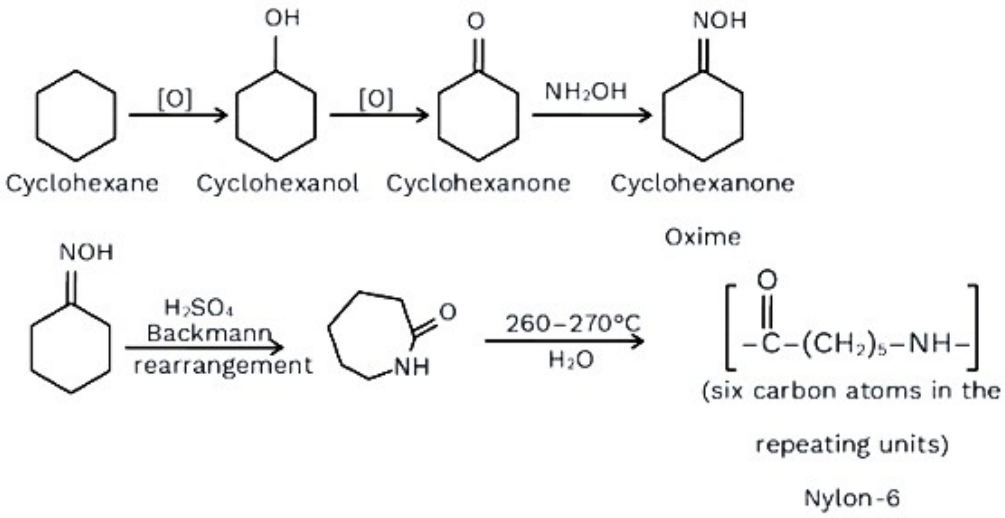
Uses of Nylon

(i) in manufacture of ropes as they are very strong ,elastic, light and flexible.
(ii) in manufacturing garments, carpets, fabrics, etc.
(iii) retain creases due to their drip dry property.
(iv) Nylon -6,10 is mixed with wool due to their inert nature towards biological and chemicals agents.

3. Nyon-6 (Perlon L)

ACaprolactam is heated with water at high temperature (260-270°C) to form Nylon-6.or in other words, self-condensation of amino caproic acid molecules is done to get this polymer. Cyclohexanone oxime undergo Backmann rearrangement

to form caprolactum. The lactam is hydrolysed by H_2O into amino acid. Accordingly, the amino acid can react with the lactam and the process goes on and onto form Perlon L as final product.



4. Nyon-2–Nylon-6

Copolymerization of glycine and amino caproic acid leads to the formation this alternating polyamide which can be degraded by microorganisms easily.

Q7 **Name the monomers units with structures for the following polymers:**
(i) Nylon-66 (ii) Neoprene

A7 **(i) Nylon-66 :** Hexamethylene diamine and Adipic acid

$H_2N-(CH_2)_6 - NH_2$, $HOOC-(CH_2)_4 - COOH$.

(ii) Neoprene : Chloroprene : $CH_2=\underset{\displaystyle Cl}{\underset{|}{C}}-CH=CH_2$

Q8 **What does the part '6, 6' mean in the name nylon-6, 6?**

A8 Hexamethylenediamine and adipic acid which contains six carbon atom in each.

Polyethylene

Types of polyethylene:

(a) Low Density Poly Ethylene (LDPE) :

Ethene undergo free radical polymerisation in the presence of traces of oxygen under high pressure of 1000-2000 atmospheres at 200°C to form low density polyethylene as product..

$$n CH_2=CH_2 \xrightarrow[1500\ atm.]{200^{o}C} \left[CH_2-CH_2\right]_n$$

Molecular mass of product formed is around 20,000 .Polyethylene has low density (0.92), moderate tensile strength, high toughness. and low melting point(110°C) . LDP is a transparent polymer having branched structure. Insulation of electricity carrying wires is done with LDP. It is also used for manufacturing packing material such as bags.

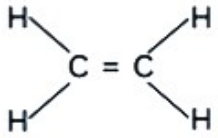
(b) High Density Poly Ethylene (HDPE) :

It is a translucent , linear chain polymer prepared in the presence of catalyst such as triethylaluminium and titanium tetrachloride at 160°C under pressure of 6 to 7 atmosphere.

It has high density (0.97) ,high melting point (130°C), tensile strength ,greater toughness, and hardness as compared to low density polyethylene. Buckets, tube, housewares, bottles are made by using HDPE.

Q9 **Draw the structure of the monomer units of each of the following polymers:**
(i) Polythene (ii) Polyvinylchloride (PVC) (iii) PMMA

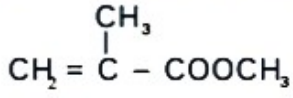
A9 (i) Monomer for the polythene is ethene



(ii) Monomer for PVC (Polyvinyl chloride) is vinyl chloride (Chloroethene)

$CH_2 = CH - Cl$

(iii) Monomer for PMMA (Polymethyl metha crylate) is methyl methacrylate



Q10 **Low density polythene obtained through :**

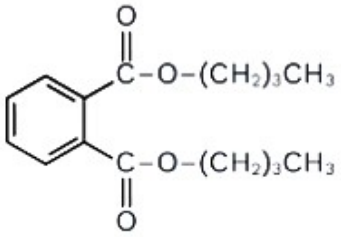
(A) Free radical addition **(B) H-atom abstraction**

(C) Both (A) and (B) **(D) None**

A10 C

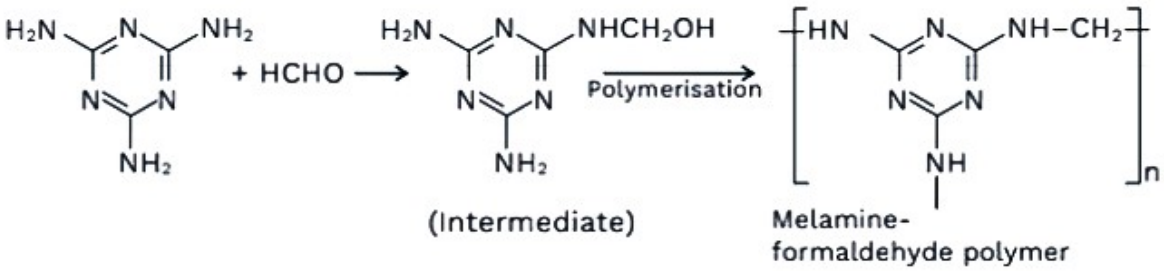
Plasticizer

A plasticiser is an organic compound that dissolves in the polymer and permits the polymer chains to slide past one another. So that the polymer becomes more flexible. Dibutylphthalate is a generally unitlized commonly used plasticiser.



Melamine - Formaldehyde Resin

Condensation polymerisation of formaldehyde and melamine results in formation of Melamine – formaldehyde resin.



The articles of melamine – formaldehyde resin are unbreakable as it is very hard. Unbreakable crockery under the name melamine are manufactured using this polymer.

Q11 **Write the**

(i) structure and

(ii) one use of Urea-formaldehyde resin.

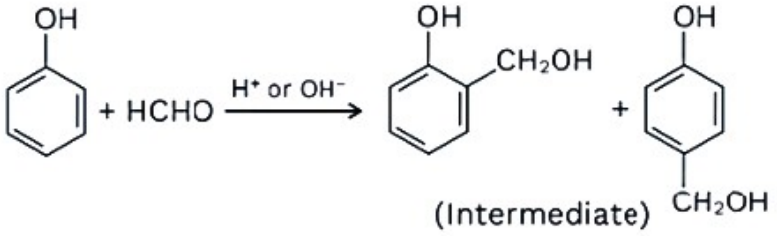
A11 **Urea-formaldehyde resin**

Structure : $\{ NH - CO - NH - CH_2 \}_n$

Use : It is used for making unbreakable cups, plates and laminated sheets.

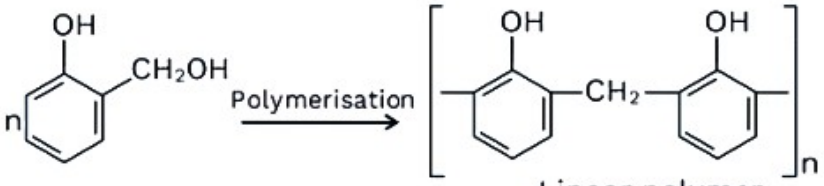
Bakelite

Phenol and formaldehyde undergo condensation reaction in the presence of catalyst such as an acid or a base. Ortho and parahydroxymethyl phenol derivatives formed in the starting react with phenol to form compounds. These compounds have rings linked to each other via –CH2 groups. The methylene bridges are formed at ortho, para or both ortho and para positions. The conditions of the reaction are responsible for formation of Straight or cross - linked materials.



(Intermediate)

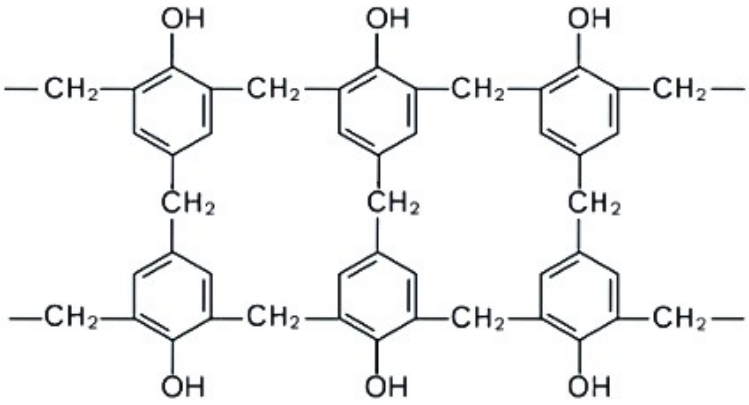
o-and p-hydroxymethyl phenol



Linear polymer

Novolac

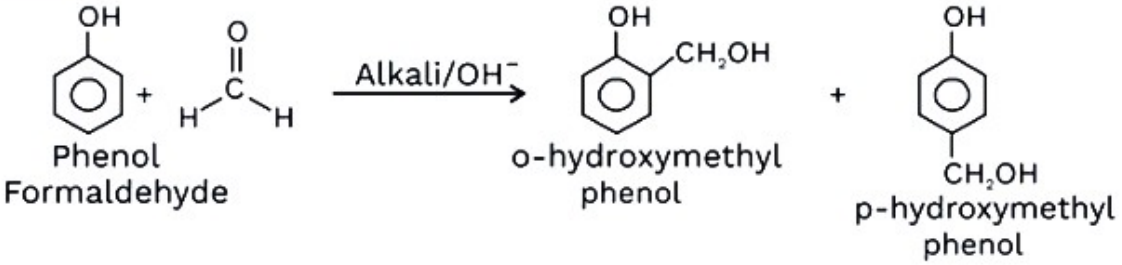
Novolac on heating with formaldehyde undergoes cross linking to form an infusible solid mass called bakelite.

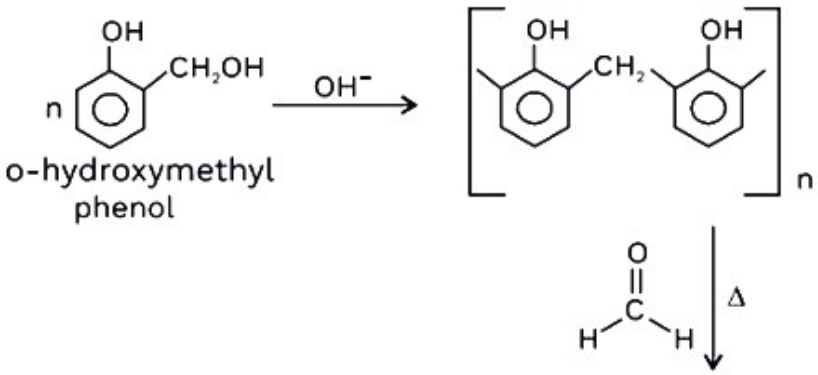


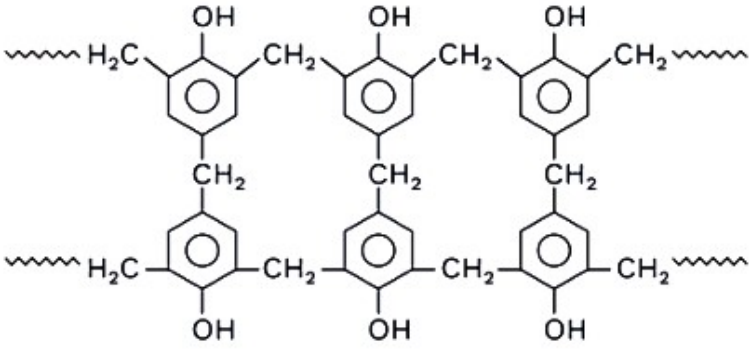
Cross - linked polymer (Bakelite)

Q12 **Discuss the synthesis of bakelite and give its use.**

A12 **Synthesis of bakelite :** Bakelite is prepared with the help of phenol and formaldehyde. Phenol and formaldehyde reacts in presence of alkali to form o-hydroxyphenol and p-hydroxyphenol then these two hydroxyphenol give bakelite.





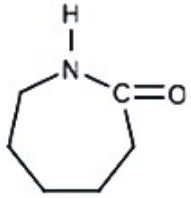


Bakelite

Uses : Soft bakelite is used as bonding for wooden planks and in the preparation of varnishes.

Q13 **Write the structures of the monomers units for the given polymers. Also draw the structures of the polymers and uses of**
(i) Nylon 6 (ii) Bakelite

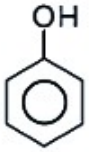
A13 **(i) Nylon 6 :**
Monomer Name : Caprolactam

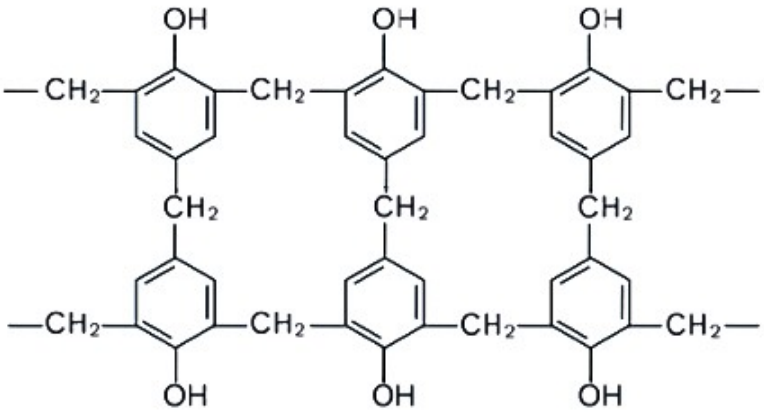
Structure : 

Structure of polymer : $(-NH-(CH_2)_5-\overset{\overset{O}{\|}}{C}-)_n$

Uses : It is used in making fibres, plastics, tyre cords and ropes.

(ii) Bakelite : (Phenol-formaldehyde resin)
Monomer Name : (a) Formaldehyde and (b) Phenol

Structure : (a) HCHO (b) C_6H_5OH i.e. 

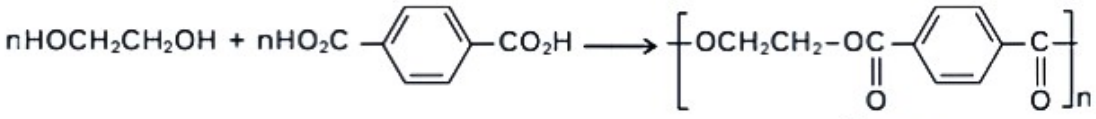
Structure of Polymer : 

Cross - linked polymer (Bakelite)

Uses : Gears, protective coating ,electrical fittingsetc., are manufactured by using bakelite.

Polyesters

Ethylene glycol is heated with terephthalic acid at 140° to 180° C in the presence of catalyst such as zinc acetate and antimony trioxide to form Dacron, polyester as final product.



Dacron

Wash and wear garments are made using this polyester because of its qualities such as crease resistant , low moisture absorption and high tensile strength.

Q14 **Write the structure of repeating monomeric units of**
(i) Dacron **(ii) Polyvinyl chloride**

A14 **(i) Dacron :** $HOH_2C - CH_2OH$ and HOOC — C$_6$H$_4$ — COOH

Ethylene glycol Terephthalic acid

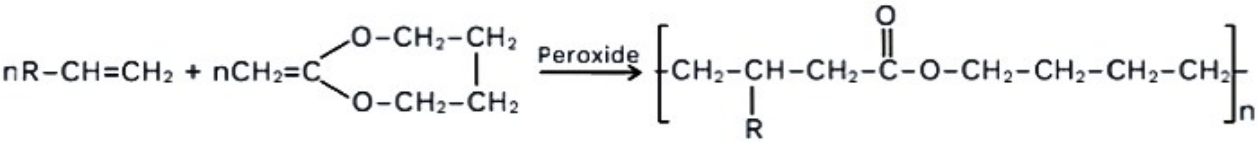
(ii) Polyvinyl chloride : $CH_2 = CH - Cl$

Vinyl chloride

Biodegradable Polymers

Plastic ,mostly used man-made polymer is non biodegradable due to unreactivity of C-C bonds of chain growth polymers toward enzyme catalysed reaction. They are serious threat to environment.

Polymers which can be degraded by catalyst easily are biodegradable polymers. The skeleton of these polymers have been designed and developed by inserting bonds that can be decomposed easily.The strategy includes insertion of functional group similar to that of biopolymers into the polymer. For instance, when acetal (I) is added during the polymerization of alkene, ester group is embedded into the polymeric chains.



Ester linkage attacked by enzyme

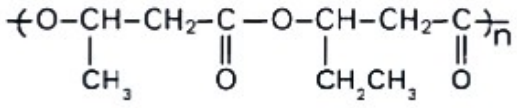
Example: Aliphatic polyesters

Other examples:

1. Poly-β-Hydroxybutyrate-CO-β-Hydroxyvalerate (PHBV)

3 hydroxypentanoic acid and 3 - hydroxybutanoic acid undergo copolymerisation to give PHBV as final product.

$$CH_3-CH(OH)-CH_2-COOH + CH_3-CH_2-CH(OH)-CH_2-COOH \longrightarrow$$



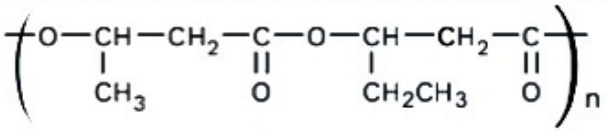
The monomeric units imparts flexibility and stiffness to PHBV. In orthopaedic devices, in controlled drug release and also for packaging purposes, this biodegradable polymer, is used. Bacterial decomposition in the environment is also done by this biodegradable polymer.

2. Polyglycolic Acid and Polylactic Acid

They contain commercially successful biodegradable polymers like sutures. The first bioadsorbable suture prepared for biodegradable polyesters for post - operative stitches was dextron.

Q15 What are biodegradable polymers? Write the structural formula of a biodegradable polymer.

A15 **Biodegradable polymers :** The natural polymers (cellulose, starch, protein, nucleic acids etc.) disintegrate by themselves during a certain period of time and hence are called biodegradable polymers. Poly-β-hydroxybutyrate-co-β-hydroxyvalerate (PHBV) is a an example of biodegradable polymer.



PHBV

Q16 Give one example of such a polymer and mention its uses.

A16 Poly-β-hydroxy butyrate-co-β-hydroxy valerate (PHBV). It is used in speciality packaging, orthopaedic devices and in controlled release of drugs.

Molecular Mass of Polymer

Naturally occurring polymers have definite molecular weight as they consists chains of same length, but for polymers having chains of varying length it is expressed as an average.

Representation of molecular mass of polymer is as follows :

(a) Number average molecular mass $\left(\overline{M}_n\right)$

$$\left(\overline{M}_n = \frac{\sum N_i M_i}{\sum_i N_i}\right)$$

where Ni = number of molecules of molecular mass Mi_i

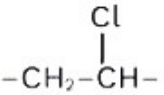
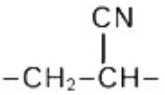
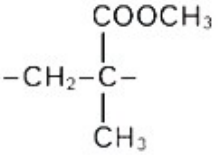
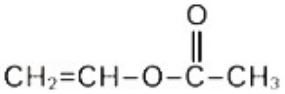
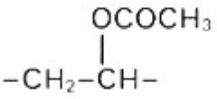
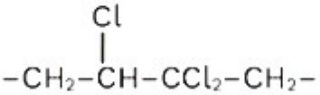
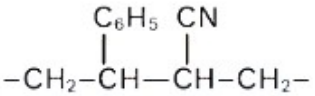
(b) Weight average molecular mass $\left(\overline{M}_n\right)$

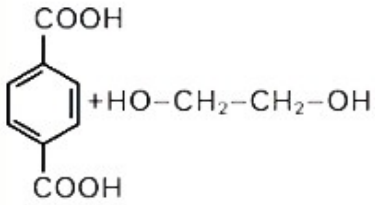
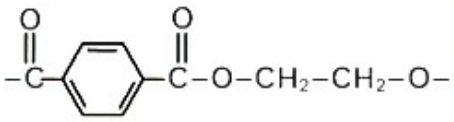
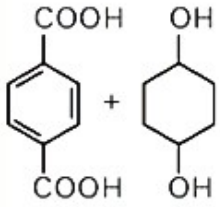
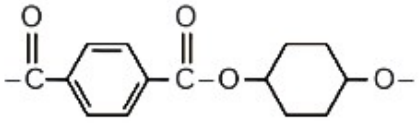
$$\overline{M}_n = \frac{\sum N_i M_i^2}{\sum_i N_i M_i}$$

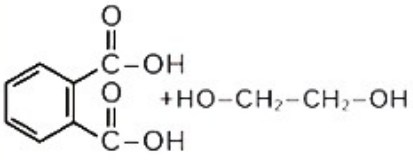
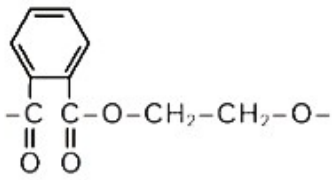
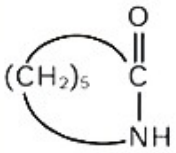
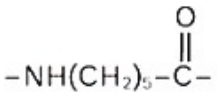
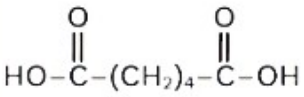
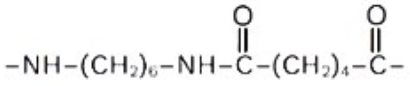
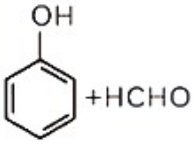
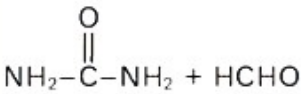
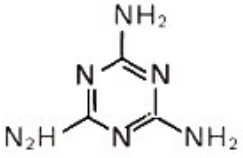
Where Ni = number of molecules of molecular mass M_i.

Common Polymers

S.No	Monomer	Repeatng unit	Polymer
1.	$CH_2=CH_2$ Ethylene	$-CH_2-CH_2-$	Polyethylene
2.	$CH_3-CH=CH_2$ Propene	$-CH_2-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-$	Polypropene
3.	$C_6H_5-CH=CH_2$ Styrene	$-CH_2-\underset{\displaystyle C_6H_5}{\underset{\vert}{CH}}-$	Polystyrene

S.No	Monomer	Repeatng unit	Polymer
4.	$CF_2=CF_2$ Tetrafluoroethylene	$-CF_2-CF_2-$	Polytetrafluoro ethylene (PTFE), Teflon
5.	$CH_2=CH-Cl$ Vinyl chloride		Polyvinyl Chloride(PVC)
6.	$CH_2=CH-CN$ Vinyl cyanide or Acrylonitrile		Polyvinyl cyanide, poly acrylonitrile, Orlon.
7.	 Methyl methacrylate		Polymethyl metha acrylate, Plexiglas, Lucite
8.	 Vinyl acetate		Polyvinyl Acetate
9.	$CH_2=CH-CH=CH_2$ 1,3-butadiene	$-CH_2-CH-CH-CH_2-$	Polybutadiene, Buna rubber
10.	$CH=CH-Cl$ (vinyl chloride) + $CH=CCl$ (Vinylidene chloride)		Saran
11.	$C_6H_5-CH=CH$ (Styrene) + $CH_2=CH-CN$ (acrylonitrile)		SAN

S.No	Monomer	Repeatng unit	Polymer
12.	$CH_2=CH-CN$ + $CH_2=CH-CH=CH_2$ + $C_6H_5-CH=CH_2$	—	ABS
13.	$CH_2=\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}-CH_3$ + $CH_2=\overset{\displaystyle CH_3}{\overset{|}{C}}-CH=CH_2$	—	Butyl rubber
14.	$C_6H_5-CH=CH_2+CH_2=CH-CH=CH_2$	—	Buna -S, SBR
15.	$CH_2=CH-CN+CH_2=CH-CH=CH_2$	—	Buna-N, NBR
16.	$CH_2=\overset{\displaystyle Cl}{\overset{|}{C}}-CH=CH_2$ Chloroprene	$-CH_2-\overset{\displaystyle Cl}{\overset{|}{C}}=CH-CH_2-$	Neoprene
17.			Polyethylene terephthalate, Terylene, Dacron or Mylar
18.			Kodel Polyster

S.No	Monomer	Repeatng unit	Polymer
19.			Polyethylene phthalate alkyd resin (Glyptal)
20.	 Caprolactam		Nylon-6
21.	$NH_2(CH_2)_6-NH_2$ Hexamethylenediamine +  Adipic Acid		Nylon – 66
22.		–	Bakelite or resol
23.		–	Urea-formaldehyde resin
24.	 Melamine	–	Malamine formaldehyde resin